



1. Description

1.1. Project

Project Name	h7FOC
Board Name	custom
Generated with:	STM32CubeMX 6.14.1
Date	03/17/2026

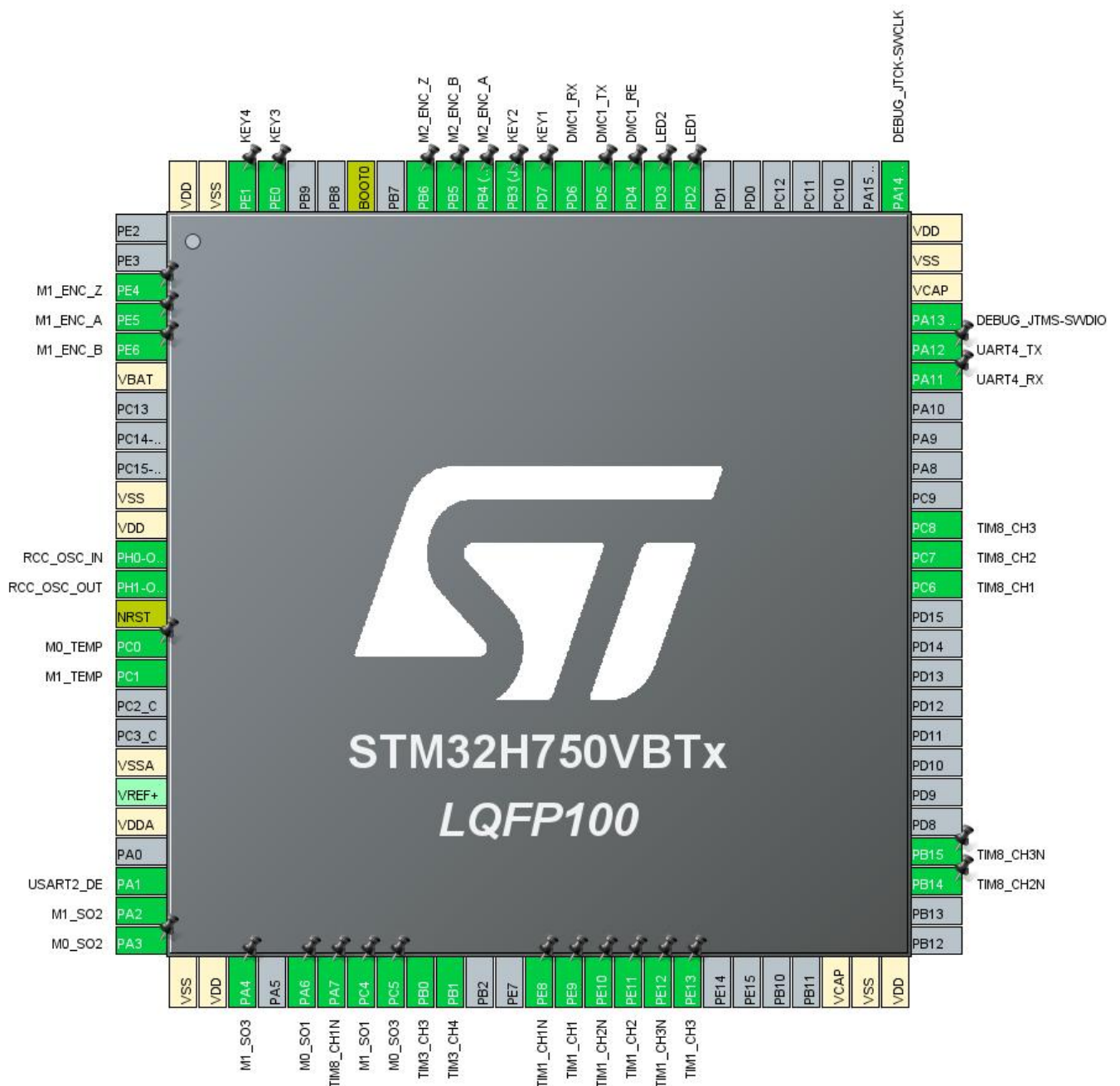
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H750 Value line
MCU name	STM32H750VBTx
MCU Package	LQFP100
MCU Pin number	100

1.3. Core(s) information

Core(s)	ARM Cortex-M7
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2. Pinout Configuration



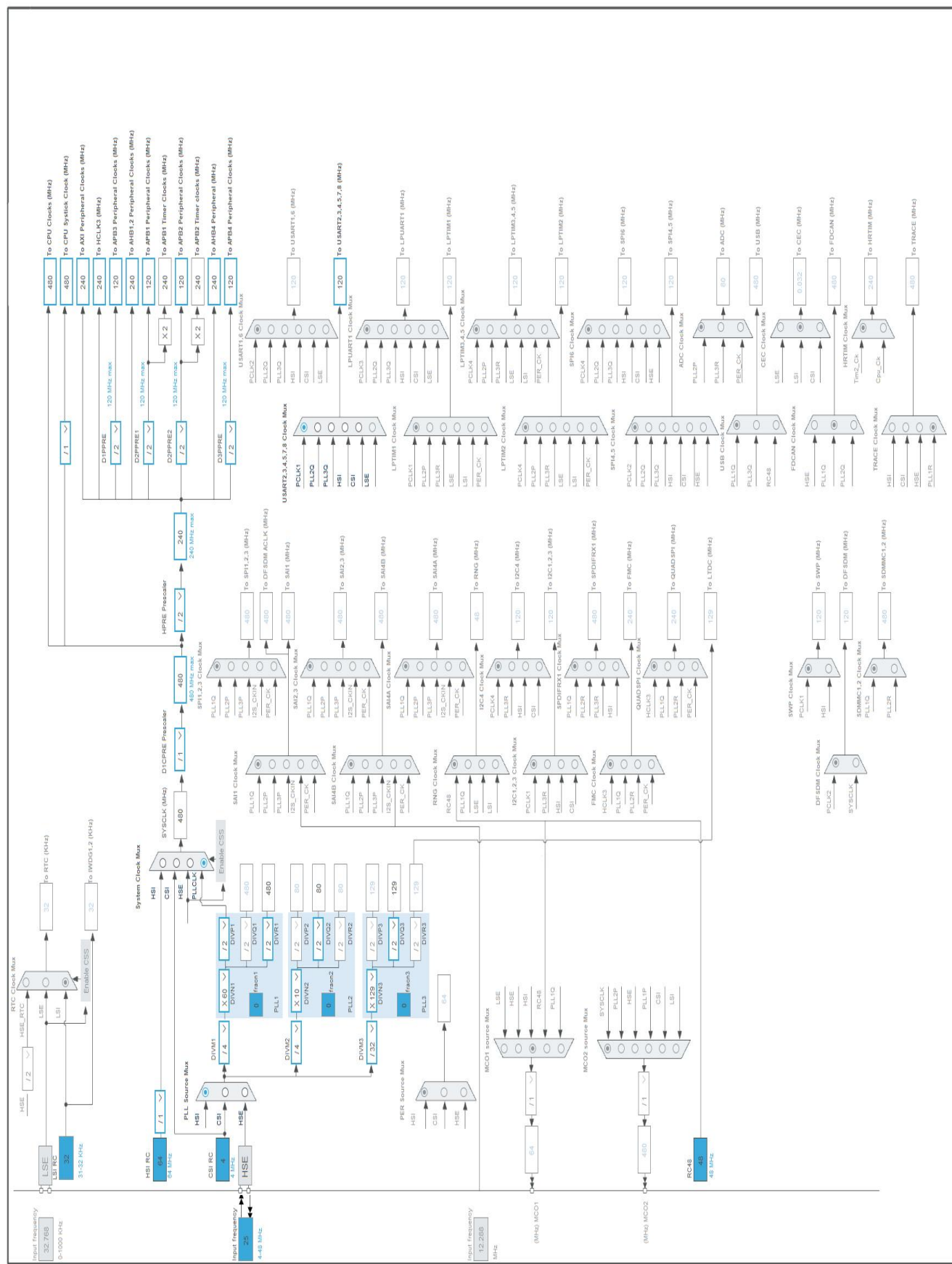
3. Pins Configuration

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
3	PE4	I/O	GPIO_EXTI4	M1_ENC_Z
4	PE5 *	I/O	GPIO_Input	M1_ENC_A
5	PE6 *	I/O	GPIO_Input	M1_ENC_B
6	VBAT	Power		
10	VSS	Power		
11	VDD	Power		
12	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
13	PH1-OSC_OUT (PH1)	I/O	RCC_OSC_OUT	
14	NRST	Reset		
15	PC0	I/O	ADC1_INP10	M0_TEMP
16	PC1	I/O	ADC1_INP11	M1_TEMP
19	VSSA	Power		
21	VDDA	Power		
23	PA1	I/O	USART2_DE	
24	PA2	I/O	ADC1_INP14	M1_SO2
25	PA3	I/O	ADC2_INP15	M0_SO2
26	VSS	Power		
27	VDD	Power		
28	PA4	I/O	ADC1_INP18	M1_SO3
30	PA6	I/O	ADC2_INP3	M0_SO1
31	PA7	I/O	TIM8_CH1N	
32	PC4	I/O	ADC1_INP4	M1_SO1
33	PC5	I/O	ADC2_INP8	M0_SO3
34	PB0	I/O	TIM3_CH3	
35	PB1	I/O	TIM3_CH4	
38	PE8	I/O	TIM1_CH1N	
39	PE9	I/O	TIM1_CH1	
40	PE10	I/O	TIM1_CH2N	
41	PE11	I/O	TIM1_CH2	
42	PE12	I/O	TIM1_CH3N	
43	PE13	I/O	TIM1_CH3	
48	VCAP	Power		
49	VSS	Power		
50	VDD	Power		
53	PB14	I/O	TIM8_CH2N	
54	PB15	I/O	TIM8_CH3N	

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
63	PC6	I/O	TIM8_CH1	
64	PC7	I/O	TIM8_CH2	
65	PC8	I/O	TIM8_CH3	
70	PA11	I/O	UART4_RX	
71	PA12	I/O	UART4_TX	
72	PA13 (JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
73	VCAP	Power		
74	VSS	Power		
75	VDD	Power		
76	PA14 (JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
83	PD2 *	I/O	GPIO_Output	LED1
84	PD3 *	I/O	GPIO_Output	LED2
85	PD4 *	I/O	GPIO_Output	DMC1_RE
86	PD5	I/O	USART2_TX	DMC1_TX
87	PD6	I/O	USART2_RX	DMC1_RX
88	PD7 *	I/O	GPIO_Input	KEY1
89	PB3 (JTDO/TRACESWO) *	I/O	GPIO_Input	KEY2
90	PB4 (NJTRST)	I/O	TIM3_CH1	M2_ENC_A
91	PB5	I/O	TIM3_CH2	M2_ENC_B
92	PB6	I/O	GPIO_EXTI6	M2_ENC_Z
94	BOOT0	Boot		
97	PE0 *	I/O	GPIO_Input	KEY3
98	PE1 *	I/O	GPIO_Input	KEY4
99	VSS	Power		
100	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H750 Value line
MCU	STM32H750VBTx
Datasheet	DS12556 Rev6

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

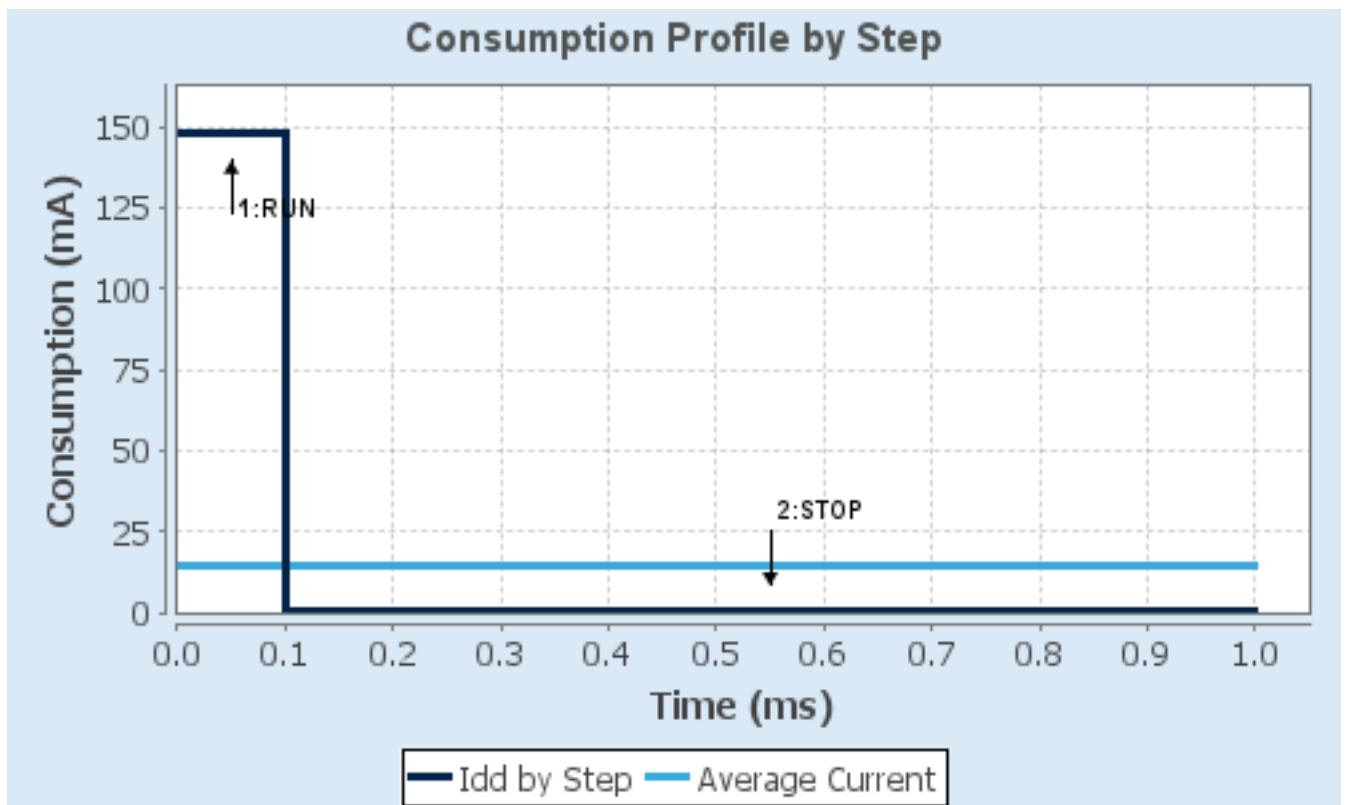
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0-High	SVOS5: System-Scale5
D1 Mode	DRUN/CRUN	DSTANDBY
D2 Mode	DRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	ITCM	NA
CPU Frequency	480 MHz	0 Hz
Clock Configuration	HSE BYP PLL	Flash-OFF
Clock Source Frequency	24 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	148 mA	150 μ A
Duration	0.1 ms	0.9 ms
DMIPS	1027.0	0.0
Ta Max	105.02	124.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	14.94 mA
Battery Life	1 day, 17 hours	Average DMIPS	1027.2001 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	h7FOC
Project Folder	C:\Users\ASUS\Desktop\STM32demo\h7FOC
Toolchain / IDE	MDK-ARM V5.32
Firmware Package Name and Version	STM32Cube FW_H7 V1.12.1
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_TIM1_Init	TIM1
5	MX_TIM8_Init	TIM8
6	MX_UART4_Init	UART4
7	MX_ADC1_Init	ADC1
8	MX_ADC2_Init	ADC2
9	MX_TIM7_Init	TIM7
10	MX_TIM15_Init	TIM15
11	MX_TIM3_Init	TIM3

Rank	Function Name	Peripheral Instance Name
12	MX_USART2_UART_Init	USART2

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN4: IN4 Single-ended

IN10: IN10 Single-ended

mode: IN11

mode: IN14

IN18: IN18 Single-ended

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler

Synchronous clock mode divided by 4 *

Resolution

ADC 16-bit resolution

Scan Conversion Mode

Enabled

Continuous Conversion Mode

Enabled *

Discontinuous Conversion Mode

Disabled

End Of Conversion Selection

End of single conversion

Overrun behaviour

Overrun data preserved

Left Bit Shift

No bit shift

Conversion Data Management Mode

DMA Circular Mode *

Low Power Auto Wait

Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable

Enable Regular Oversampling

Disable

Oversampling Ratio

1

Number Of Conversion

3 *

External Trigger Conversion Source

Timer 1 Trigger Out event *

External Trigger Conversion Edge

Trigger detection on the rising edge

Rank

1

Channel

Channel 11 *

Sampling Time

64.5 Cycles *

Offset Number

No offset

Offset Signed Saturation

Disable

Rank

2 *

Channel

Channel 4

Sampling Time

1.5 Cycles

Offset Number

No offset

Offset Signed Saturation	Disable
<u>Rank</u>	3 *
Channel	Channel 18 *
Sampling Time	1.5 Cycles
Offset Number	No offset
Offset Signed Saturation	Disable
ADC_Injected_ConversionMode:	
Enable Injected Conversions	Enable *
Enable Injected Oversampling	Disable
Number Of Conversions	2 *
External Trigger Source	Timer 1 Trigger Out event *
External Trigger Conversion Edge	Trigger detection on the rising edge
Injected Conversion Mode	None
Injected Queue	Injected Queue Disable
<u>Rank</u>	1
Channel	Channel 10 *
Sampling Time	64.5 Cycles *
Offset Number	No offset
Offset Signed Saturation	Disable
Injected Offset Saturation	Disable
<u>Rank</u>	2 *
Channel	Channel 11 *
Sampling Time	64.5 Cycles *
Offset Number	No offset
Offset Signed Saturation	Disable
Injected Offset Saturation	Disable
Analog Watchdog 1:	
Enable Analog WatchDog1 Mode	false
Analog Watchdog 2:	
Enable Analog WatchDog2 Mode	false
Analog Watchdog 3:	
Enable Analog WatchDog3 Mode	false

3.2. ADC2

IN3: IN3 Single-ended
mode: IN8
mode: IN15

3.2.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler

Synchronous clock mode divided by 4 *

Resolution

ADC 16-bit resolution

Scan Conversion Mode

Enabled

Continuous Conversion Mode

Disabled

Discontinuous Conversion Mode

Disabled

End Of Conversion Selection

End of single conversion

Overrun behaviour

Overrun data preserved

Left Bit Shift

No bit shift

Conversion Data Management Mode

DMA Circular Mode *

Low Power Auto Wait

Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable

Enable Regular Oversampling

Disable

Oversampling Ratio

1

Number Of Conversion

3 *

External Trigger Conversion Source

Timer 1 Trigger Out event 2 *

External Trigger Conversion Edge

Trigger detection on the rising edge

Rank

1

Channel

Channel 15 *

Sampling Time

64.5 Cycles *

Offset Number

No offset

Offset Signed Saturation

Disable

Rank

2 *

Channel

Channel 3

Sampling Time

64.5 Cycles *

Offset Number

No offset

Offset Signed Saturation

Disable

Rank

3 *

Channel

Channel 8 *

Sampling Time

64.5 Cycles *

Offset Number

No offset

Offset Signed Saturation

Disable

ADC_Injected_ConversionMode:

Enable Injected Conversions

Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode false

3.3. CORTEX_M7

3.3.1. Parameter Settings:

Speculation default mode Settings:

Speculation default mode **Enabled ***

Cortex Interface Settings:

CPU ICache Disabled

CPU DCache Disabled

Cortex Memory Protection Unit Control Settings:

MPU Control Mode Background Region Privileged accesses only + MPU Disabled during hard fault,
NMI and FAULTMASK handlers

Cortex Memory Protection Unit Region 0 Settings:

MPU Region Enabled

MPU Region Base Address **0x0 ***

MPU Region Size 4GB

MPU SubRegion Disable **0x87 ***

MPU TEX field level level 0

MPU Access Permission ALL ACCESS NOT PERMITTED

MPU Instruction Access DISABLE

MPU Shareability Permission ENABLE

MPU Cacheable Permission DISABLE

MPU Bufferable Permission DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 2 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 3 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 4 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 5 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 6 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 7 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 8 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region Disabled

3.4. DEBUG

Debug: Serial Wire

3.5. MEMORYMAP

mode: Activated

3.6. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.6.1. Parameter Settings:

Power Parameters:

SupplySource	PWR_LDO_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

RCC Parameters:

TIM Prescaler Selection	Disabled
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HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
CSI Calibration Value	32
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	4 WS (5 CPU cycle)
Product revision	rev.V

PLL range Parameters:

PLL1 clock Input range	Between 8 and 16 MHz
PLL1 clock Output range	Wide VCO range

3.7. SYS

Timebase Source: TIM6

3.8. TIM1

Channel1: PWM Generation CH1 CH1N

Channel2: PWM Generation CH2 CH2N

Channel3: PWM Generation CH3 CH3N

Channel4: PWM Generation No Output

3.8.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	3 *
Counter Mode	Center Aligned mode1 *
Counter Period (AutoReload Register - 16 bits value)	2999 *
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	1 *
auto-reload preload	Enable *

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Output Compare (OC4REF) *

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0

BRK Sources Configuration

- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0

BRK2 Sources Configuration

- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Enable *
Off State Selection for Run Mode (OSSR)	Enable *
Off State Selection for Idle Mode (OSSI)	Enable *
Lock Configuration	Off
Dead Time	100 *

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1 and 1N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 2 and 2N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 3 and 3N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 4:

Mode	PWM mode 1
Pulse (16 bits value)	1500 *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.9. TIM3

Channel3: Input Capture direct mode

Channel4: Input Capture direct mode

Combined Channels: Encoder Mode

3.9.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Enable *

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Input Capture Channel 3:

Polarity Selection	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter (4 bits value)	0

Input Capture Channel 4:

Polarity Selection	Rising Edge
IC Selection	Direct

Prescaler Division Ratio	No division
Input Filter (4 bits value)	0

Encoder:

Encoder Mode

Encoder Mode TI1 and TI2 *

____ Parameters for Channel 1 ____

Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

____ Parameters for Channel 2 ____

Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.10. TIM7

mode: Activated

3.10.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	239 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	999 *
auto-reload preload	Enable *

Trigger Output (TRGO) Parameters:

Trigger Event Selection	Reset (UG bit from TIMx_EGR)
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3.11. TIM8

Channel1: PWM Generation CH1 CH1N

Channel2: PWM Generation CH2 CH2N

Channel3: PWM Generation CH3 CH3N

3.11.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	

2999 *

Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- DFSDM	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off
Dead Time	0

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1 and 1N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset

CHN Idle State	Reset
PWM Generation Channel 2 and 2N:	
Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 3 and 3N:	
Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

3.12. TIM15

mode: Clock Source

3.12.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 8 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

3.13. UART4

Mode: Asynchronous

3.13.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.14. USART2

Mode: Asynchronous

mode: Hardware Flow Control (RS485)

3.14.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable

ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration
Polarity	High
Assertion Time	0
Deassertion Time	0

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.15. FREERTOS

Interface: CMSIS_V2

3.15.1. Config parameters:

API:

FreeRTOS API	CMSIS v2
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Versions:

FreeRTOS version	10.3.1
CMSIS-RTOS version	2.00

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Disabled

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled

USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	15360
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME	Enabled
USE_OS2_THREAD_ENUMERATE	Enabled
USE_OS2_EVENTFLAGS_FROM_ISR	Enabled
USE_OS2_THREAD_FLAGS	Enabled

USE_OS2_TIMER	Enabled
USE_OS2_MUTEX	Enabled

3.15.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Enabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.15.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled
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Project settings (see parameter description first):

Use FW pack heap file	Enabled
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*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC0	ADC1_INP10	Analog mode	No pull-up and no pull-down	n/a	M0_TEMP
	PC1	ADC1_INP11	Analog mode	No pull-up and no pull-down	n/a	M1_TEMP
	PA2	ADC1_INP14	Analog mode	No pull-up and no pull-down	n/a	M1_SO2
	PA4	ADC1_INP18	Analog mode	No pull-up and no pull-down	n/a	M1_SO3
	PC4	ADC1_INP4	Analog mode	No pull-up and no pull-down	n/a	M1_SO1
ADC2	PA3	ADC2_INP15	Analog mode	No pull-up and no pull-down	n/a	M0_SO2
	PA6	ADC2_INP3	Analog mode	No pull-up and no pull-down	n/a	M0_SO1
	PC5	ADC2_INP8	Analog mode	No pull-up and no pull-down	n/a	M0_SO3
DEBUG	PA13 (JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
	PA14 (JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
RCC	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT (PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
TIM1	PE8	TIM1_CH1N	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE9	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE10	TIM1_CH2N	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE11	TIM1_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE12	TIM1_CH3N	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE13	TIM1_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM3	PB0	TIM3_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB1	TIM3_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB4 (NJTRST)	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	M2_ENC_A
	PB5	TIM3_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	M2_ENC_B
TIM8	PA7	TIM8_CH1N	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB14	TIM8_CH2N	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB15	TIM8_CH3N	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC6	TIM8_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC7	TIM8_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC8	TIM8_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
UART4	PA11	UART4_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PA12	UART4_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART2	PA1	USART2_DE	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD5	USART2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	DMC1_TX
	PD6	USART2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	DMC1_RX
GPIO	PE4	GPIO_EXTI4	External Interrupt Mode with Rising edge trigger detection	Pull-up *	n/a	M1_ENC_Z
	PE5	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	M1_ENC_A
	PE6	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	M1_ENC_B
	PD2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED1
	PD3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED2
	PD4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DMC1_RE
	PD7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	KEY1
	PB3 (JTDO/TRACESWO)	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	KEY2
	PB6	GPIO_EXTI6	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	M2_ENC_Z
	PE0	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	KEY3
	PE1	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	KEY4

4.2. DMA configuration

DMA request	Stream	Direction	Priority
ADC1	DMA1_Stream1	Peripheral To Memory	Low
UART4_RX	DMA1_Stream0	Peripheral To Memory	Low
UART4_TX	DMA1_Stream2	Memory To Peripheral	Low
ADC2	DMA1_Stream3	Peripheral To Memory	Low
USART2_RX	DMA1_Stream4	Peripheral To Memory	Low
USART2_TX	DMA1_Stream5	Memory To Peripheral	Low

ADC1: DMA1_Stream1 DMA request Settings:

Mode: **Circular ***
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Half Word
 Memory Data Width: Half Word

UART4_RX: DMA1_Stream0 DMA request Settings:

Mode: **Circular ***
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

UART4_TX: DMA1_Stream2 DMA request Settings:

Mode: **Circular ***
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

ADC2: DMA1_Stream3 DMA request Settings:

Mode: **Circular ***
Use fifo: Disable
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Half Word
Memory Data Width: Half Word

USART2_RX: DMA1_Stream4 DMA request Settings:

Mode: **Circular ***
Use fifo: Disable
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

USART2_TX: DMA1_Stream5 DMA request Settings:

Mode: **Circular ***
Use fifo: Disable
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

4.3. BDMA configuration

nothing configured in DMA service

4.4. MDMA configuration

nothing configured in DMA service

4.5. NVIC configuration

4.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
EXTI line4 interrupt	true	3	0
DMA1 stream0 global interrupt	true	6	0
DMA1 stream1 global interrupt	true	5	0
DMA1 stream2 global interrupt	true	5	0
DMA1 stream3 global interrupt	true	1	0
DMA1 stream4 global interrupt	true	5	0
DMA1 stream5 global interrupt	true	5	0
ADC1 and ADC2 global interrupts	true	2	0
EXTI line[9:5] interrupts	true	3	0
TIM3 global interrupt	true	0	0
UART4 global interrupt	true	6	0
TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts	true	15	0
TIM7 global interrupt	true	2	0
DMAMUX1 overrun interrupt	true	5	0
PVD and AVD interrupts through EXTI line 16		unused	
Flash global interrupt		unused	
RCC global interrupt		unused	
TIM1 break interrupt		unused	
TIM1 update interrupt		unused	
TIM1 trigger and commutation interrupts		unused	
TIM1 capture compare interrupt		unused	
USART2 global interrupt		unused	
TIM8 break interrupt and TIM12 global interrupt		unused	
TIM8 update interrupt and TIM13 global interrupt		unused	
TIM8 trigger and commutation interrupts and TIM14 global interrupt		unused	
TIM8 capture compare interrupt		unused	
FPU global interrupt		unused	

Interrupt Table	Enable	Preenmption Priority	SubPriority
TIM15 global interrupt		unused	
HSEM1 global interrupt		unused	

4.5.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
EXTI line4 interrupt	false	true	true
DMA1 stream0 global interrupt	false	true	true
DMA1 stream1 global interrupt	false	true	true
DMA1 stream2 global interrupt	false	true	true
DMA1 stream3 global interrupt	false	true	true
DMA1 stream4 global interrupt	false	true	true
DMA1 stream5 global interrupt	false	true	true
ADC1 and ADC2 global interrupts	false	true	true
EXTI line[9:5] interrupts	false	true	true
TIM3 global interrupt	false	true	true
UART4 global interrupt	false	true	true
TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts	false	true	true
TIM7 global interrupt	false	true	true
DMAMUX1 overrun interrupt	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

5.1.2. Without filters

5.2. Power Domain view

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7_bsd1.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
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Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
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Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
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Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4502-stm32-smbuspmibus-expansion-package-for-stm32cube-stmicroelectronics.pdf
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Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6088-how-to-use-mce-for-encryptiondecryption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6179-how-to-integrate-the-stl-firmware-into-a-time-critical-user-application-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf
Design Notes & Tips	https://www.st.com/resource/en/design_tip/dt0117-microphone-array-beamforming-in-the-pcm-and-pdm-domain-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0392-stm32h742xig-stm32h743xig-stm32h750xb-stm32h753xi-device-errata-stmicroelectronics.pdf

Datasheet	https://www.st.com/resource/en/datasheet/dm00496677.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0253-stm32f7-series-and-stm32h7-series-cortexm7-processor-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0433-stm32h742-stm32h743753-and-stm32h750-value-line-advanced-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf
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